

Pairs Trading, Empirical Validation II

Single Stocks, Intraday Data, Pre-registered Thresholds and Korean Data

Re-tested across three markets and four frequencies. Three results went against expectations.

Stocks

Intraday

Pre-reg

Korea

5,886 US stock pairs · 1-minute bars · 8 Korean preferred pairs — all real data

EXECUTIVE SUMMARY

The core finding held; the other three were overturned

All four items listed on the last slide of Empirical Validation I were tested.

Replicated A 2σ threshold in every market and frequency; the reversion rate rises 1.5–1.9× in each

Flipped ① Intraday data still cannot identify information shares — 1-min residual corr. 0.899, needed 0.20

Flipped ② A pre-registered threshold still lost to conventional $\pm 2\sigma$. Per-pair estimation error eats the gain

Flipped ③ Korean preferred pairs are untradable. With an 81-day half-life, borrowing is half the cost

2σ is not the true model — it is the optimal shrinkage estimate of the population threshold

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Four sessions and what each closes

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Single stocks

109 stocks, 5,886 pairs re-tested

S8

Intraday data

1-min, 5-min, 1-hour bars

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A true walk-forward test

S10

Korean real data

8 common-preferred pairs

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Corrections

Which claims were wrong

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Six lines

S7

Stocks

What changes when the universe grows 30×

109 US large caps · 5,886 pairs · 2,949 trading days

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S7. Single stocks

The baseline is the 13% placebo, not the nominal 5%

Same test on random walks with each stock's drift and volatility. Counts are of the 5,239 pairs tested.

$$\Pr(\text{reject} \mid H_0)_{\text{nominal}} = 0.05 \longrightarrow \widehat{\text{size}}_{\text{placebo}} = 0.132$$

Baseline	Pass rate	Pairs	Reading
Nominal level	5.0%	262	Naive baseline
Placebo test (109 random walks)	13.2%	692	Correct baseline
Real data (ADF, lag 1)	34.5%	1,807	2.6× placebo — real signal

Three placebo runs: 10.4 · 15.7 · 13.6% — drift makes the ADF over-reject

S7. Single stocks

The economic rationale of sectors added no information

Contrary to the survey literature, there is no statistical edge.

Group	Pairs	Passed	Pass rate	Mean OOS Sharpe
Within sector	715	247	34.55%	0.19
Across sectors	4,524	1,560	34.48%	0.20

Possible reading ①

- Sector classes are too coarse
- Needs business similarity, not GICS level

Possible reading ②

- Large caps already share common factors
- Sector adds nothing on top

Measure, don't trust — check in every sample whether candidate pruning works

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Intraday

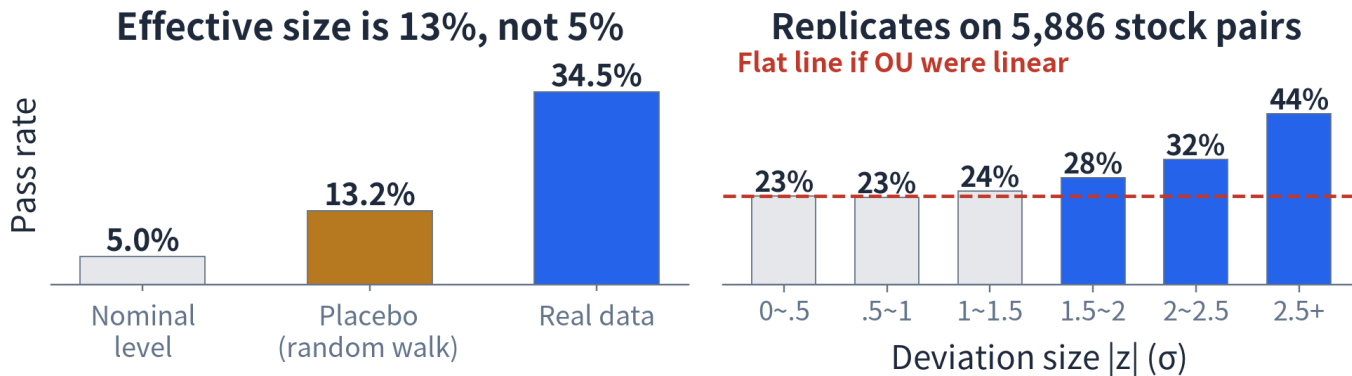
Pre-reg

Korea

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S7. Single stocks

The core finding replicates in a 300× larger sample



13.2%

Placebo effective size
nominal is 5%

23→44%

Reversion rate range
1.9×

26→49%

ETF sample
1.9× — same

1.36M

Observations
498-pair sample

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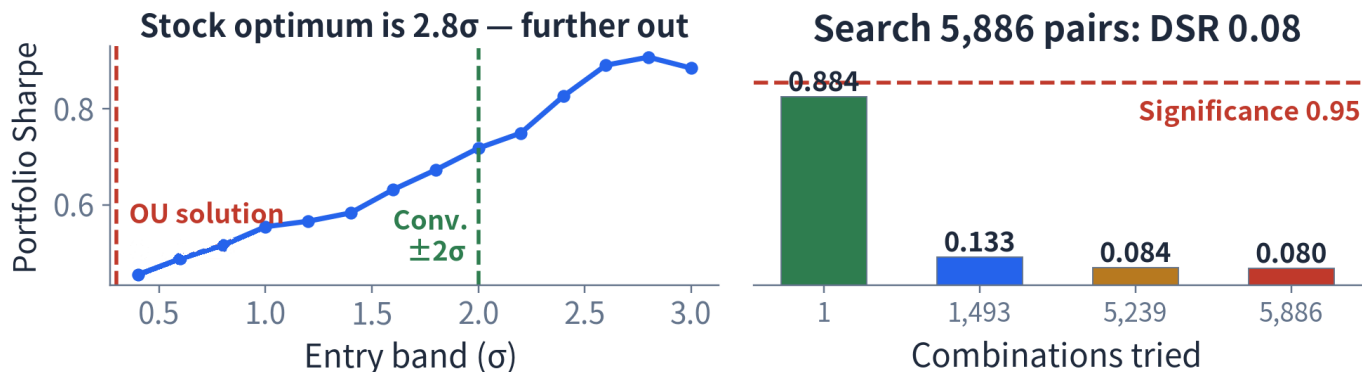
Pre-reg

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S7. Single stocks

| Grow the search space 30x and the DSR collapses

**2.8 σ** **Empirical optimal band**ETF: 2.2σ **0.70****Portfolio Sharpe**

1,493 pairs, equal-weight

64%**In→out decay**

0.55 → 0.20

0.080**DSR (5,886 trials)**

threshold 0.95

S8

Intraday

Do 1-min bars identify info shares?

Epps effect · bound width · threshold by frequency · time of day

S8. Intraday

The bound width is a function of residual correlation

Higher frequency lowers the correlation — the Epps effect.

$$\rho_{\Delta t} = \text{Corr}(r_{t,\Delta t}^{(1)}, r_{t,\Delta t}^{(2)}) \xrightarrow{\Delta t \downarrow} \rho_{\infty} < \rho_{\text{daily}}$$

$$IS^{\max} - IS^{\min} = \frac{(n_1\sigma_1 + n_2\rho\sigma_2)^2 - n_1^2\sigma_1^2(1-\rho^2)}{n_1^2\sigma_1^2 + 2n_1n_2\rho\sigma_1\sigma_2 + n_2^2\sigma_2^2}$$

- **Validity check** The decomposition holds only if the two adjustment coefficients have opposite signs
- **Report width** If the width exceeds 0.3, claim no ranking
- **Invert** Solve for the correlation a target width needs; compare with the measured value

Only the lag count scales with frequency; the rest of the pipeline is unchanged

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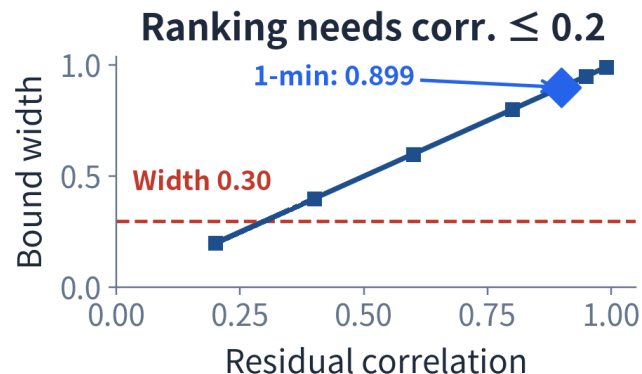
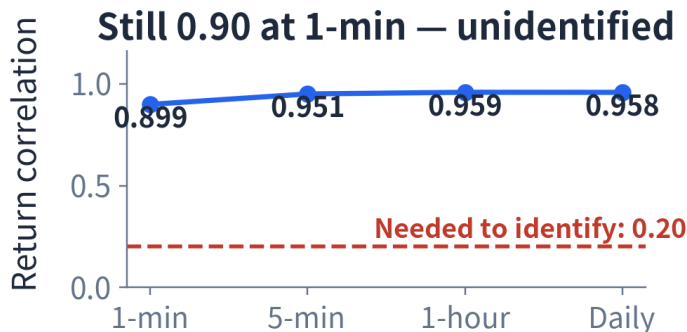
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S8. Intraday

Still unidentified even with 1-minute bars

**0.899**

1-min residual corr.
daily 0.958

0.20

Required level
for width 0.3

0.83

1-min mean width
pairs with width < 0.3: 0%

4/9

Decomposition holds
daily 0/9

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S8. Intraday

Correcting the Empirical Validation I claim

Empirical Validation I said intraday data would solve it.

Empirical Validation I claim

- Information share is an intraday-data problem
- Higher frequency narrows the bounds
- → get intraday data and it is solved

Empirical Validation II test

- 1-min residual corr. 0.899 — not enough
- Width 0.3 needs corr. ≤ 0.20
- → must go down to ticks and quotes

In practice

Without tick data, the data cannot say which leg leads price discovery. Use the liquidity ranking instead

Overview

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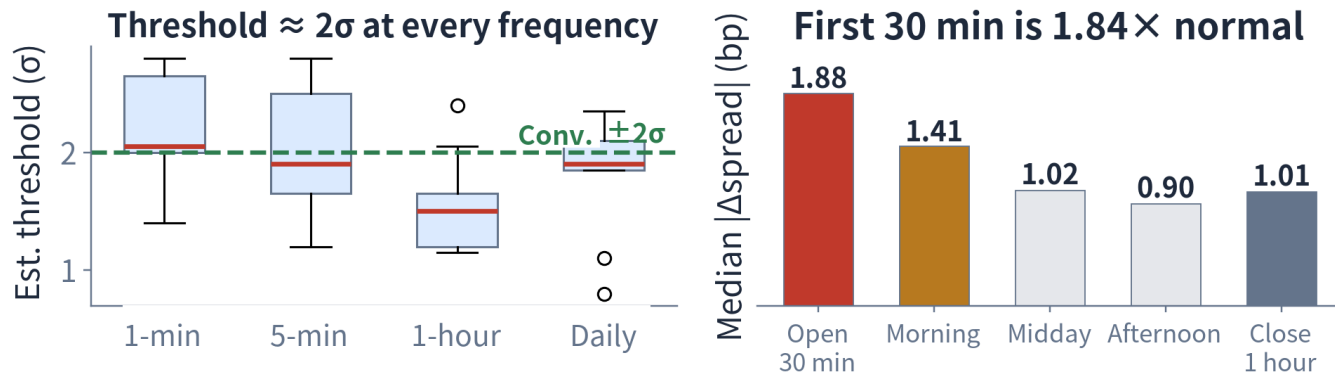
Intraday

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S8. Intraday

The threshold is near 2σ at every frequency **2.05σ** 1-min threshold
range 1.40–2.80 **1.90σ** 5-min threshold
most stable **1.90σ** Daily threshold
Validation I: 1.88σ **$1.84\times$** First 30 min
spread volatility

S9

Pre-reg

Does fixing the threshold in advance win?

In-sample estimate · out-of-sample freeze · shrinkage sweep

S9. Pre-registration

Estimate in-sample only, freeze out-of-sample

Sweep the shrinkage weight from 0 to 1 and look for an interior optimum.

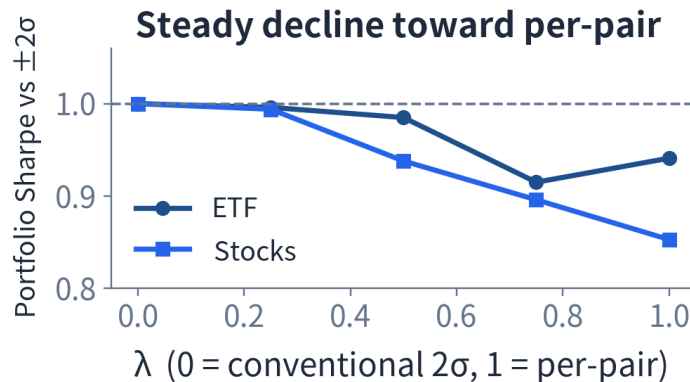
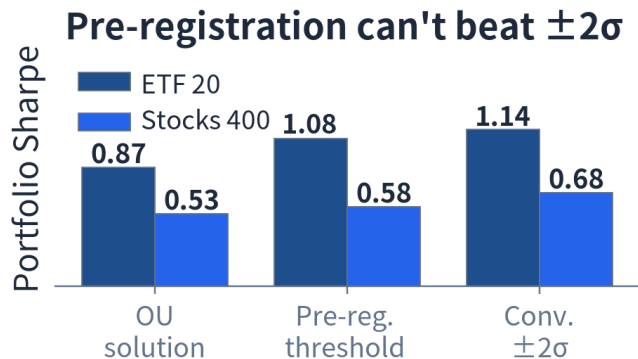
$$\hat{\theta}_i = \arg \min_{\theta} \text{SSR}_i(\theta) \Big|_{\text{in-sample}} \implies b_i^{\text{OOS}} \equiv \hat{\theta}_i$$

$$b_i(\lambda) = \lambda \hat{\theta}_i + (1 - \lambda) \bar{b}, \quad \bar{b} = 2.0\sigma$$

- **Three rules** Per-pair threshold · conventional $\pm 2\sigma$ · OU solution, on the same data
- **Two metrics** Pair Sharpe and portfolio Sharpe can rank differently
- **DSR included** Trying three rules also counts as trials

It is pre-registered only if not a single out-of-sample observation was seen

S9. Pre-registration

Pre-registered, it still did not beat conventional $\pm 2\sigma$ **1.08**ETF threshold
 $\pm 2\sigma$: 1.14**0.58**Stock threshold
 $\pm 2\sigma$: 0.68**40~42%**Pairs that won
a coin flip **$\lambda = 0$** Optimal shrinkage
keep the convention

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S9. Pre-registration

The population value was right; per-pair deviation was noise

The median estimated threshold, $1.80\text{--}1.85\sigma$, is almost exactly 2σ .

1 Evidence ①

The interquartile range is wide, $1.45\text{--}2.20\sigma$, yet that spread is not signal

2 Evidence ②

Corr. of (threshold $- 2\sigma$) with (performance gap): $+0.544$ for 20 ETF pairs, $+0.040$ for 400 stock pairs

3 Reading

The relation vanished as the sample grew; $+0.544$ on 20 pairs was small-sample luck

4 Same structure

The same estimation-error problem as S2, where mean-variance lost to equal weights

Conventional 2σ is a shrinkage estimator of the population threshold — and that shrinkage was optimal

S10

Korea

Are Korean preferred-share pairs tradable?

2,876 days of real data · 8 common-preferred pairs · 2026 costs

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S10. Korea

The simulation in Pairs Trading in Korea missed badly

The real common-preferred gap is far larger and far slower.

Item	Korea deck assumption	Actual	Ratio
Mean gap	15.0%	49.1%	3.3×
Std. dev.	2.50%p	5.38%p	2.2×
Half-life	25 days	81 days (median)	3.2×
ADF rejects	—	2/8 pairs	cannot confirm
Effective sample $\kappa \cdot T$	baseline 30	13 (median)	below baseline

The preferred discount is not a mean-reverting shock but a slow-moving governance variable

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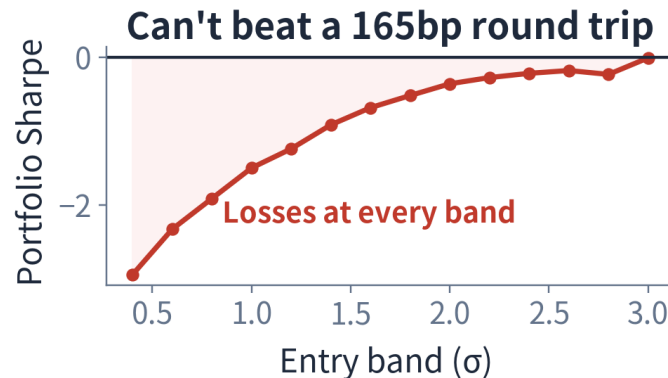
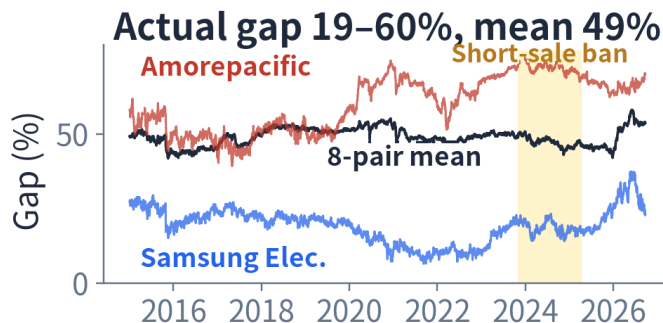
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S10. Korea

| Out-of-sample losses at every band

**165bp**

Round-trip cost

0.31 σ of sd**81bp**

Borrow share

half the cost

81 days

Half-life

borrow fees accrue

3.0 σ

Best band

still Sharpe 0.00

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S10. Korea

Slow mean reversion and borrow costs do not mix

Round-trip cost with the 2026 sell-side tax of 0.20%.

$$c_{\text{rt}}^{\text{KR}} = 4f + 2\tau_s + 2(i_p + i_c) + \ell \frac{h}{250}, \quad \tau_s = 20 \text{ bp (2026)}$$

Item	bp	Share
Long leg (buy/sell preferred)	53.0	32%
Short leg (borrow-sell/cover common)	31.0	19%
Stock-borrow fee (2.5% p.a. × 81-day half-life)	80.9	49%
Round-trip total	164.9	100%

The slower the reversion, the longer you pay to borrow — the half-life sets the cost

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S10. Korea

During the short-selling ban, the gap actually narrowed

Counter to intuition — but do not read it causally.

Period	Days	Mean gap	Change in period	Trend
Before ban (2021–2023.11)	699	49.1%	−3.5%p	+0.2%p/yr
Ban (2023.11–2025.3)	339	47.4%	−3.0%p	−3.1%p/yr
After resumption (2025.4–)	364	49.1%	+6.8%p	+8.5%p/yr

Confounders

The Value-up Program and buyback debate overlapped the same period, and a common-share-led rally may have widened the gap mechanically

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S10. Korea

Even for index ETFs, cost is 5.6× the spread volatility

KODEX 200 and TIGER 200 track the same index.

9.1bp**Log-spread sd**

250-day mean removed

1.4 days**Half-life**

ADF p 0.0000

51bp**Round-trip cost**

incl. 20bp sell tax

5.63 σ **Cost / sd**

no entry possible

The statistics are perfect

- ADF p 0.0000, half-life 1.4 days
- No doubt about stationarity

Still not tradable

- No deviation large enough to enter
- Same structure as the cash-futures (basis) result in Pairs Trading in Korea

Summary

Summary

What held, and what must be corrected

Three confirmed · four claims to correct · six recommendations

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The 2σ threshold held in three markets and four frequencies

Sample	Frequency	Est. threshold	Reversion-rate rise
US ETFs, 20 pairs	Daily	1.88σ	26 → 49% (1.9×)
US stocks, 5,886 pairs	Daily	2.8σ (sweep)	23 → 44% (1.9×)
US ETFs, 9 pairs	1-min	2.05σ	—
US ETFs, 9 pairs	5-min	1.90σ	—
US ETFs, 9 pairs	1-hour	1.50σ	—
Korean preferred, 8 pairs	Daily	—	33 → 48% (1.5×)

The no-arbitrage band is not an artefact of asset class or sampling frequency — it is structural

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Summary

Correcting four Empirical Validation I claims

Empirical Validation I claim	Empirical Validation II test
Information share is an intraday-data problem	1-min bars are not enough — corr. 0.899, needed 0.20
Conventional $\pm 2\sigma$ was close to the true model	2σ is a shrinkage estimator of the population threshold, and the shrinkage is optimal
Pre-register a threshold-based band	Tried it; it lost to $\pm 2\sigma$ — per-pair estimation error eats the gain
Korean preferred pairs (simulated)	Untradable on real data — 81-day half-life, borrowing is half the cost

Write down the limits, and the next test aims straight at them

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Six practical recommendations

Each comes directly from this evidence.

- **Band** Fix it at 2σ . Adjust by asset class if at all, never per pair
- **Baseline** Measure the effective size with a placebo test first. Do not use the nominal 5%
- **Universe** Decide expansion together with the DSR: 190 pairs 0.667 → 5,886 pairs 0.080
- **Time of day** Hold signals for the first 30 minutes; spread volatility is $1.84\times$
- **Half-life** Read it with the borrow cost; slow reversion is eaten by borrowing
- **Info share** Without tick data, do not attempt it. Use the liquidity ranking instead

Five of the six are about testing, costs and constraints — not signals

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Summary

Settings needed to reproduce

Session	Data	Period
S7	109 US large caps, daily adjusted close	2015-01-02 – 2026-09-24 (2,949 days)
S8	15 ETFs, 1-min / 5-min / 1-hour bars	5 days / 60 days / 2 years
S9	20 ETF pairs + 400 stock pairs	Selection 2015–2020, evaluation 2021–2026
S10	8 Korean common-preferred pairs, 2 index ETFs	2015-01-02 – 2026-09-23 (2,876 days)

Cost assumptions

US round trip 5bp · Korea round trip 164.9bp (2026 sell-side tax 0.20%) · rolling 90-day z-score

Only the placebo test and the required-correlation inversion are simulated; everything else is real data

EMPIRICAL VALIDATION II

One thing replicated; three had to be corrected

The habit of writing down limits becomes the table of contents of the next test